2.3.3 Costs and benefits

In this subsection, we will take a closer look at the costs and benefits of the logs signal type. Table 2.1 shows an overview of the costs.

Table 2.1 Overview of the costs of logs

	Overhead	Considerations
Instrumentation	Small and simple to add a log line	Widely adopted and supported by programming languages
Agent	Some runtime, lots of options, and few standards for formats (syslog, OTLP)	A runtime footprint, ingestion performance, reliability, and security
Destination	Long-term storage can be expensive.	Data temperature (access recency or frequency) can guide retention

INSTRUMENTATION COSTS OF LOGS

The costs of logs are relatively low for instrumentation, well understood, and widely supported. All programming languages and their libraries allow users to emit logs. Readability, and with it maintainability, has not greatly suffered by adding log lines.

Let's have a concrete look at the instrumentation costs from a developer's point of view. The original code in listing 2.1 has 24 lines of code, and the code instrumented with logs in listing 2.3 has 32 lines of code. Taking the two LOC numbers together, we can calculate the B2I ratio and arrive at a value of 1.33, which is not too bad.

TELEMETRY COSTS OF LOGS

There are some costs associated with collecting and routing logs to a central location, which is typically necessary in a distributed system. Depending on the option you are using here, it may cause ingestion (network traffic) costs.

When selecting a certain agent, ask yourself the following:

- What is the runtime footprint (CPU and memory consumption)?
- What is the ingestion performance? For example, how many log lines per second can the agent handle?
- In terms of reliability, is there a guaranteed delivery of every log line (if this is necessary at all)?
- Is there an appropriate level of security, from encryption on the wire (TLS) to multitenancy support?

TIP If you're unsure about how to measure log agent performance, have a look at chapter 7 of Phil Wilkins' *Logging in Action* (Manning, 2022; https://www.manning.com/books/logging-in-action).

The following are a few examples of log routers: